Task 1 — Foundation & Environment Setup

Cybersecurity & Ethical Hacking Internship | ApexPlanet Software Pvt. Ltd.

Intern	Aryan Doshi
Duration	Days 1–12
Date	2026-06-07
Status	Completed

Objective

Build strong fundamentals in cybersecurity, networking, and cryptography, and set up a professional penetration testing lab using VirtualBox, Kali Linux, and Metasploitable2. This forms the foundation for all subsequent internship tasks.

Lab Environment Setup

Component	Details	Status
Host Machine	Windows PC — VirtualBox 7.x	Installed
Attacker Machine	Kali Linux (VirtualBox Image)	Running
Target Machine	Metasploitable2 (.vmdk)	Running
Network Type	Host-Only Adapter (Private Lab Network)	Configured
Kali IP	192.168.56.xxx	Active
Metasploitable2 IP	192.168.56.xxx	Active

Setup Steps

- 1. Downloaded and installed VirtualBox 7.x on Windows host
- 2. Imported Kali Linux OVA image into VirtualBox
- 3. Imported Metasploitable2 .vmdk as a new VM
- 4. Set both VMs to Host-Only Adapter for isolated lab network
- 5. Verified connectivity with ping between both machines

Cybersecurity Basics

CIA Triad

Principle	Description
Confidentiality	Ensuring only authorized users can access data
Integrity	Ensuring data is not tampered with or altered

Availability	Ensuring systems and data are accessible when needed

Threat Types Explored

Threat	Description
Phishing	Fake emails/websites designed to steal credentials
Malware	Malicious software — viruses, trojans, spyware
DDoS	Distributed Denial of Service attacks to crash servers
SQL Injection	Injecting malicious SQL into database queries
Brute Force	Systematically trying all possible passwords
Ransomware	Encrypts victim files and demands payment

Attack Vectors Studied

- Social Engineering — Manipulating people to reveal confidential information
- Wireless Attacks — Targeting Wi-Fi networks (Evil Twin, Deauth attacks)
- Insider Threats — Attacks originating from within an organization

Linux Fundamentals Practiced

Category	Commands Used	Purpose
File Navigation	cd, ls, pwd	Navigate the file system
Permissions	chmod, chown	Manage file/directory access
Package Mgmt	apt, dpkg	Install/update software
Networking	ifconfig, ping, netstat, traceroute	Network diagnostics

See Linux_Cheatsheet.pdf for the full command reference.

Networking Basics

- **OSI Model** — Studied all 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application
- **TCP/IP Suite** — How TCP/IP protocols power the internet and local networks
- **DNS & HTTP/HTTPS** — How domain names resolve to IPs; how web traffic is encrypted with TLS
- **IP Addressing** — Subnetting, CIDR notation, and NAT concepts

OSI Model Quick Reference

Layer	Name	Example Protocols
7	Application	HTTP, FTP, DNS, SMTP
6	Presentation	SSL/TLS, JPEG, ASCII
5	Session	NetBIOS, RPC

4	Transport	TCP, UDP
3	Network	IP, ICMP, ARP
2	Data Link	Ethernet, MAC, Wi-Fi
1	Physical	Cables, Hubs, Signals

Cryptography Basics

Topic	Description
Symmetric Encryption	Same key for encryption & decryption (e.g., AES-256)
Asymmetric Encryption	Public/Private key pair (e.g., RSA-2048)
Hashing	MD5, SHA256 — one-way data transformation for integrity
SSL/TLS	Digital certificates securing HTTPS web traffic

Hands-On: OpenSSL in Kali Linux

```
# Generate a private key
openssl genrsa -out private.key 2048
# Encrypt a message (symmetric AES-256)
openssl enc -aes-256-cbc -salt -in plaintext.txt -out encrypted.bin
# Decrypt the message
openssl enc -aes-256-cbc -d -in encrypted.bin -out decrypted.txt
# Generate SHA256 hash
echo "hello world" | openssl dgst -sha256
# Generate MD5 hash
echo "hello world" | openssl dgst -md5
```

Tool Familiarization

Tool	Purpose	Status
Wireshark	Packet capture and network traffic analysis	Explored
Nmap	Network scanning — port and service detection	Used
Burp Suite	Web proxy for intercepting HTTP/HTTPS traffic	Familiarized
Netcat	Network debugging and connection testing	Familiarized

Nmap Scan Results on Metasploitable2

```
nmap -sV 192.168.56.xxx
```

Port	Service	Version	Risk
21/TCP	FTP	vsftpd 2.3.4	CRITICAL

22/TCP	SSH	OpenSSH 4.7p1	**Medium**
23/TCP	Telnet	Linux telnetd	**High**
80/TCP	HTTP	Apache 2.2.8	**High**
139/445/TCP	SMB	Samba 3.x	**CRITICAL**
3306/TCP	MySQL	MySQL 5.0.51a	**High**
5432/TCP	PostgreSQL	PostgreSQL 8.3	**Medium**
8180/TCP	HTTP	Apache Tomcat	**High**

Deliverables Completed

Deliverable	Status	Details
Lab Setup Report	Done	PDF with screenshots
GitHub Repository	Done	Notes & Linux cheat-sheet
5-min Video Walkthrough	Done	Screen recording of lab setup
Wireshark Test Capture	Done	Captured network packets
Nmap Scan	Done	Scanned Metasploitable2

What's Next — Task 2

Network Security & Scanning (Days 13–24)

- Reconnaissance (Whois, Shodan, Google Dorking)
- Port & Service Scanning with Nmap
- Vulnerability Scanning with OpenVAS/Nessus
- Packet Analysis with Wireshark
- Firewall basics with iptables

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